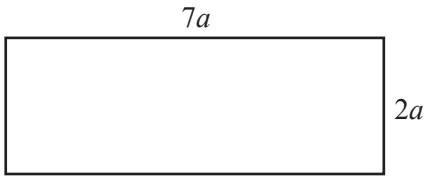
0580_s19_qp_31_Q5

5 **(a)** The diagram shows a rectangle with length $7a$ and width $2a$.



NOT TO SCALE

Write an expression, in its simplest form, for

(i) the perimeter,

..... [2]

(ii) the area.

..... [2]

(b) The nth term of a sequence is $n^2 + 5$.

Find the first three terms of this sequence.

..... , ... , ... [2]

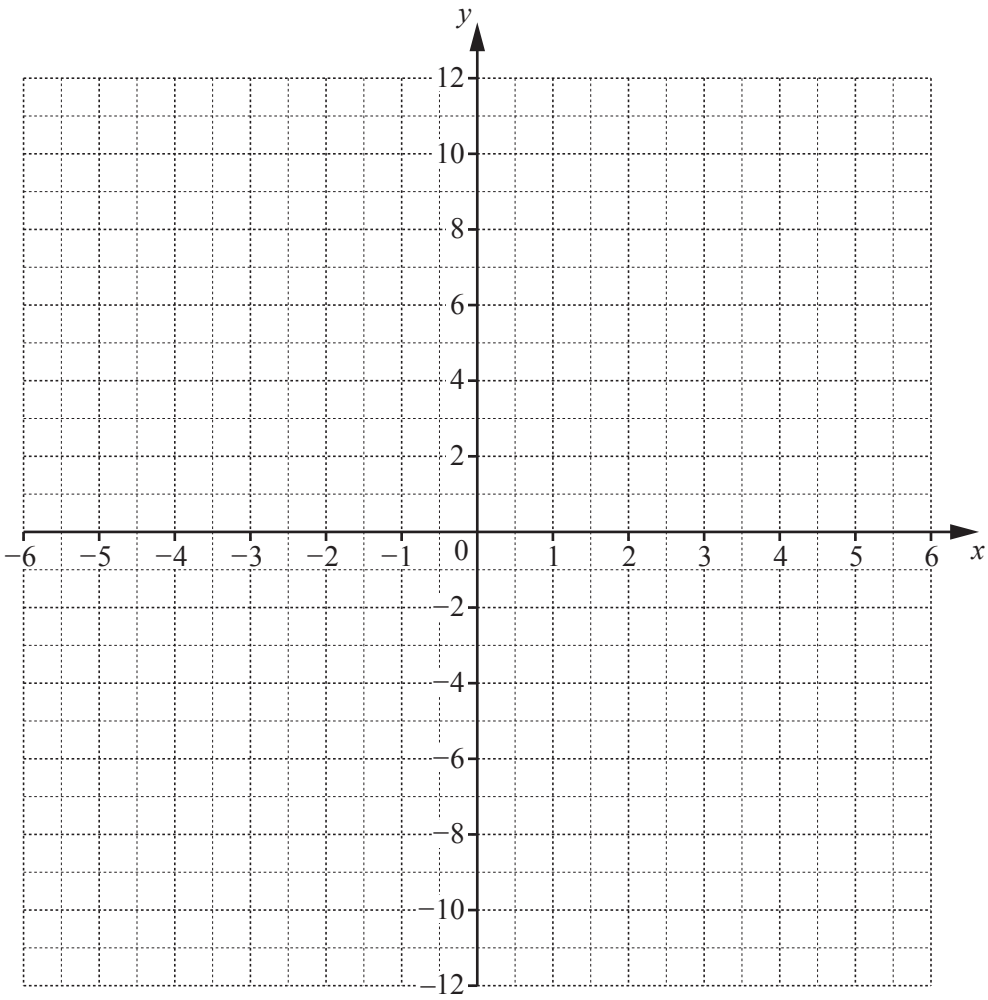
0580_s19_qp_31_Q5

(c) **(i)** Complete the table of values for $y = \frac{12}{x}$, $x \neq 0$.

x	-6	-4	-3	-2	-1		1	2	3	4	6
y	-2	-3					12				2

[3]

(ii) On the grid, draw the graph of $y = \frac{12}{x}$ for $-6 \leqslant x \leqslant -1$ and $1 \leqslant x \leqslant 6$.



[4]

(iii) On the grid, draw the line $y = 8$. [1]

(iv) Use your graph to solve $\frac{12}{x} = 8$.

$x =$ ..... [1]